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Subject : Problem Solving Using Python

Batch : 03

Mini Project: Student Record System using CSV

AIM

To develop a Python program that performs file operations to store, retrieve, search, and delete student records using a CSV file.

ALGORITHM

1. Start the program
2. Create a CSV file if it does not exist
3. Display menu options:
    - Add Student
    - View Students
    - Search Student
    - Delete Student
    - Exit
4. Ask user to choose an option
5. If choice = Add:
    - Get name, age, course
    - Append data to CSV file
6. If choice = View:
    - Read all records from file
    - Display them
7. If choice = Search:

    - Get student name
    - Search in file
    - Display if found
8. If choice = Delete:
    - Get student name
    - Remove that record from file
9. Repeat until user exits
10. Stop the program

CODE

```
import csv
import os
FILE_NAME = "students.csv"
def create_file():
    if not os.path.exists(FILE_NAME):
        with open(FILE_NAME, "w", newline="") as file:
            writer = csv.writer(file)
            writer.writerow(["Name", "Age", "Course"])

def add_student():
    name = input("Enter Name: ").title()
    age = input("Enter Age: ")
    course = input("Enter Course: ").title()

    with open(FILE_NAME, "a", newline="") as file:
        writer = csv.writer(file)
        writer.writerow([name, age, course])

    print("Student added successfully!")

def view_students():
    with open(FILE_NAME, "r") as file:
        reader = csv.reader(file)
        print("\n--- Student Records ---")
        for row in reader:
            print(row)
```

```
def search_student():
    name = input("Enter name to search: ").title()
    found = False

    with open(FILE_NAME, "r") as file:
        reader = csv.reader(file)
        for row in reader:
            if row and row[0] == name:
                print("Record Found:", row)
                found = True
                break

    if not found:
        print("Student not found")

def delete_student():
    name = input("Enter name to delete: ").title()
    rows = []

    with open(FILE_NAME, "r") as file:
        reader = csv.reader(file)
        rows = list(reader)

    with open(FILE_NAME, "w", newline="") as file:
        writer = csv.writer(file)
        for row in rows:
            if row and row[0] != name:
                writer.writerow(row)

    print("Record deleted if existed")

def main():
    create_file()

    while True:
        print("\n1. Add Student")
        print("2. View Students")
        print("3. Search Student")
        print("4. Delete Student")
        print("5. Exit")
```

```
    choice = input("Enter choice: ")

    if choice == '1':
        add_student()
    elif choice == '2':
        view_students()
    elif choice == '3':
        search_student()
    elif choice == '4':
        delete_student()
    elif choice == '5':
        break
    else:
        print("Invalid choice")

main()
```

SAMPLE OUTPUT

```
C:\Users\Nikash> python student_project.py

1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Exit

Enter choice: 1
Enter Name: Arun
Enter Age: 19
Enter Course: AIDS
Student added successfully!

1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Exit

Enter choice: 1
Enter Name: Ravi
Enter Age: 20
Enter Course: CSE
Student added successfully!

1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Exit

Enter choice: 2

--- Student Records ---
['Name', 'Age', 'Course']
['Arun', '19', 'AIDS']
['Ravi', '20', 'CSE']

1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Exit
```

```
Enter choice: 3
Enter name to search: Arun
Record Found: ['Arun', '19', 'AIDS']

1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Exit

Enter choice: 4
Enter name to delete: Ravi
Record deleted if existed

1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Exit

Enter choice: 2

--- Student Records ---
['Name', 'Age', 'Course']
['Arun', '19', 'AIDS']

1. Add Student
2. View Students
3. Search Student
4. Delete Student
5. Exit

Enter choice: 5
```

RESULT

The program was successfully executed. Student records were stored, retrieved, searched, and deleted using a CSV file. Thus, file handling operations were implemented effectively in Python.